

High Efficiency Broadband Polarization Converter Based on Tapered Slot Waveguide

Yonghao Fei, Libin Zhang, Tongtong Cao, Yanmei Cao, and Shaowu Chen

Abstract—A novel high efficiency broadband polarization converter based on an adiabatic tapered vertical slot waveguide is proposed and analyzed. Unlike previously reported tapered polarization converters, this device converts the first-order TM/TE mode into the first-order TE/TM mode. The structure utilizes a vertical slot waveguide, which only needs a one-mask etching process. The bandwidth is ultrawide, ~ 100 nm (1500–1600 nm) with a conversion efficiency of $> 98\%$. The total length of the polarization converter is $116\ \mu\text{m}$, and its minimum fabrication tolerance is acceptable for modern nanofabrication technology.

Index Terms—Optical interconnects, photonic integrated circuits, polarization converter, slot waveguide.

I. INTRODUCTION

SILICON-ON-INSULATOR (SOI) platform is promising for the photonic integrated circuits (PICs) due to its high-index-contrast (HIC) and compatibility with the complementary metal–oxide–semiconductor (CMOS) technology. However, the birefringence induced by HIC is difficult to eliminate. Therefore, many solutions have been proposed, including the polarization-independent waveguide [1], polarization splitter/converter [2]–[11] and polarization diversity system [12]–[14]. Polarization converters (PCs), as a fundamental device in PIC, have been reported previously many times, including slanted cores [6]–[8] and partially-etched cores [9]–[11]. However, the fabrication processes for the above mentioned PCs are very difficult and complex, where the gray lithography, wet-etching or multi-etching processes are utilized. Thus, PCs based on the tapered waveguide, which could reduce the fabrication complexity, has been proposed [15]–[19]. So far, the reported taper-based polarization converters deal with the polarization mode conversion between the first-order TM mode and high-order TE modes, which means that the waveguide must be multi-mode and therefore is impossible for low-loss single mode propagation. D. Dai *et al.* [19] once added a directional coupler at the end of the converter to transform the second-order TE mode to the first-order TE mode for maintaining the single-mode nature at the output waveguide.

Manuscript received December 24, 2012; revised March 10, 2013; accepted March 19, 2013. Date of publication March 27, 2013; date of current version April 16, 2013. This work was supported by the National Natural Science Foundation of China under Grant 60877013 and Grant 61021003.

The authors are with the State Key Laboratory on Optoelectronics, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China (e-mail: feiyonghao@semi.ac.cn; zhanglibin@semi.ac.cn; ttcao@semi.ac.cn; ymeicao@semi.ac.cn; swchen@semi.ac.cn).

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Digital Object Identifier 10.1109/LPT.2013.2254706

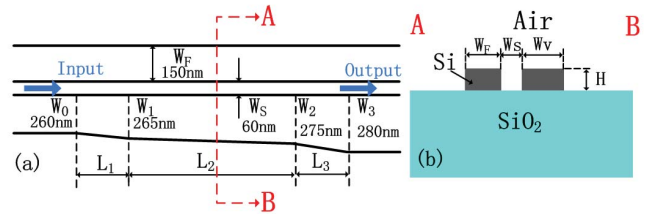


Fig. 1. Schematic representation of the polarization converter based on the asymmetric slot waveguide. (a) Top view. (b) Cross-sectional view.

The slot waveguide [20] is a special waveguide with a low-refractive-index region sandwiched between two high-refractive-index regions, which can enhance the light intensity in the slot region due to the large discontinuity of the electric field at high index-contrast interfaces. Such unique feature implies that the effective mode for the vertical (horizontal) slot waveguide can only be the TE(TM) mode, which makes it necessary to convert the TM/TE mode to TE/TM mode. The horizontal slot waveguide mode converter has been reported [17], which rotates the TM input by 90 degree while maintains the TE input. In addition, the mode transformer between the strip waveguide and the slot waveguide has been successfully demonstrated in experiment with the efficiency as high as 97% [21], [22], which shows the feasibility of the integration of the strip and slot waveguides.

In this letter, we propose a polarization converter based on a tapered vertical slot waveguide. The device can convert the first-order TM/TE mode to the first-order TE/TM mode without any other additional structures needed. The bandwidth is about 100 nm (1500 nm–1600 nm) with the conversion efficiency more than 98% and the total length is $116\ \mu\text{m}$. Furthermore, only one lithography mask is needed in the fabrication process, thus reducing the fabrication complexity significantly.

II. STRUCTURE DESIGN AND WORKING PRINCIPLE

The design is based on an asymmetric slot waveguide, which consists of a straight and a tapered silicon strip, and its top and cross-section views are shown in Fig. 1. The height of the waveguide (H) is 250 nm, while the width of the slot region (W_S) and the straight silicon strip (W_F) are 60 nm and 150 nm, respectively. The refractive indices of the Si and SiO_2 are 3.476 and 1.444, respectively. The structure breaks the symmetry in the vertical and horizontal directions, leading to the mode conversion. We use the commercial software Fimmwave [23] to calculate the effective index of the waveguide at 1550 nm

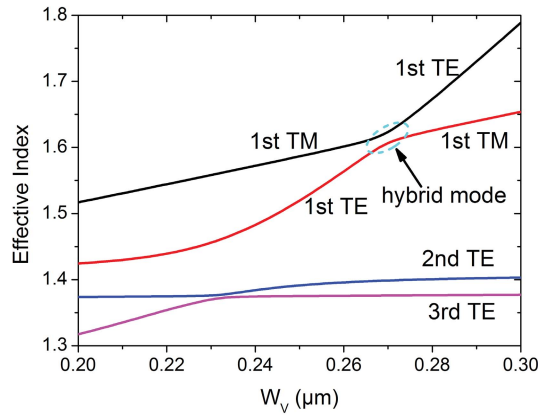


Fig. 2. Slot waveguide effective index as a function of the taper width (W_V).

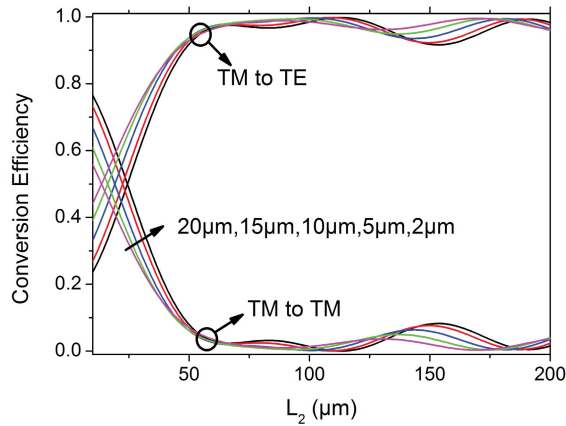


Fig. 3. Mode conversion efficiencies as a function of the taper length L_2 when the taper length L_1 and $L_3 = 2, 5, 10, 15$, and $20 \mu\text{m}$, respectively.

wavelength. As the width of the tapered strip increases, the effective index also increases, which is shown in Fig. 2. When the taper width approaches 270 nm , the effective indices of the TE and TM modes converge to a point, a bit larger than 1.6 . In other words, there exists mode hybridization near the width of 270 nm , which induces the mode conversion [15], [16]. Based on the above analysis, we propose a structure based on the principle of mode hybridization in the tapered slot waveguide.

In order to guarantee the single mode operation and sufficient mode-hybridization, we choose the start (W_0) and end width (W_3) of the tapered silicon strip to be 260 nm and 280 nm , respectively. Meanwhile, for the sake of compactness, we adopt a three-segment taper, which was proposed in [19], as shown in Fig. 1a with all the structural parameters illustrated. The taper angle of the second taper should keep small enough to make sure that it is adiabatic. The end widths (W_1 , and W_2) for the second segment should be chosen in the domain around $W = 270 \text{ nm}$, where the mode conversion happens. In the present design, we choose $W_1 = 265 \text{ nm}$ and $W_2 = 275 \text{ nm}$. The taper length L_2 should be optimized to achieve maximized mode-conversion efficiency. The length of the first segment L_1 and the third segment L_3 is ultra-short to reduce the total size of the device. In order to simplify the design, we choose the same length for the first and third taper segments ($L_1 = L_3$).

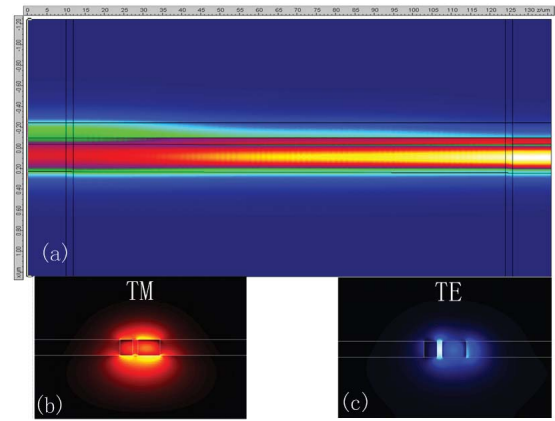


Fig. 4. (a) Light intensity propagation with the input-TM mode. Electrical field distribution of (b) input-TM mode and (c) output-TE mode.

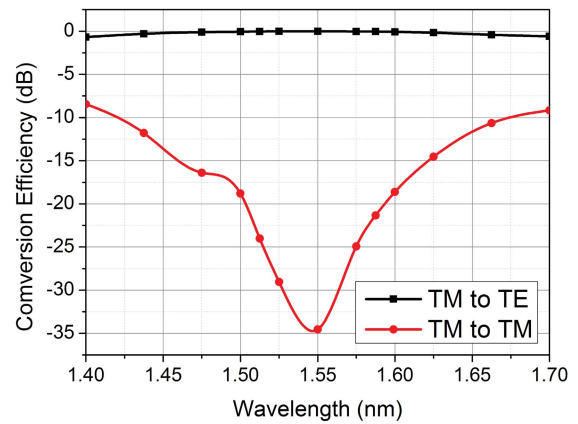


Fig. 5. Wavelength dependence of the polarization converter.

In the following part, we use the commercial software Fimmprop [23] to simulate the light propagation in the structure. Fig. 3 illustrates the mode conversion efficiencies for the input-TM mode when both the taper length L_1 and L_3 equal to $2 \mu\text{m}$, $5 \mu\text{m}$, $10 \mu\text{m}$, $15 \mu\text{m}$ and $20 \mu\text{m}$, respectively. It is clear that the conversion efficiency is more than 90% as long as the second taper length L_2 is larger than $50 \mu\text{m}$. In order to achieve higher efficiency and lower polarization crosstalk, we choose L_1 and L_3 to be $2 \mu\text{m}$, and $L_2 = 112 \mu\text{m}$. The conversion efficiency is 99.78% with the crosstalk about -35 dB at 1550 nm . Meanwhile, we can see some notable ripples in the curves, and the oscillation is weaker when L_1 and L_3 are longer, which is mentioned in [19] as well.

The propagation of light intensity for the input-TM mode is shown in Fig. 4a. As the light propagates, the mode is converted from TM mode to the hybrid mode first, and then to TE mode. An almost perfect TE mode is obtained at the end of the converter with a conversion efficiency of 99.78% . Furthermore, the electric field distribution of both the input-TM and output-TE modes are shown in Fig. 4b and 4c, which is confined loosely around the slot waveguide for TM mode, but tightly in the slot region for TE mode.

We also give the wavelength dependence of the polarization converter which is shown in Fig. 5. It can be seen that the crosstalk is lower than -18 dB for the operating wavelength

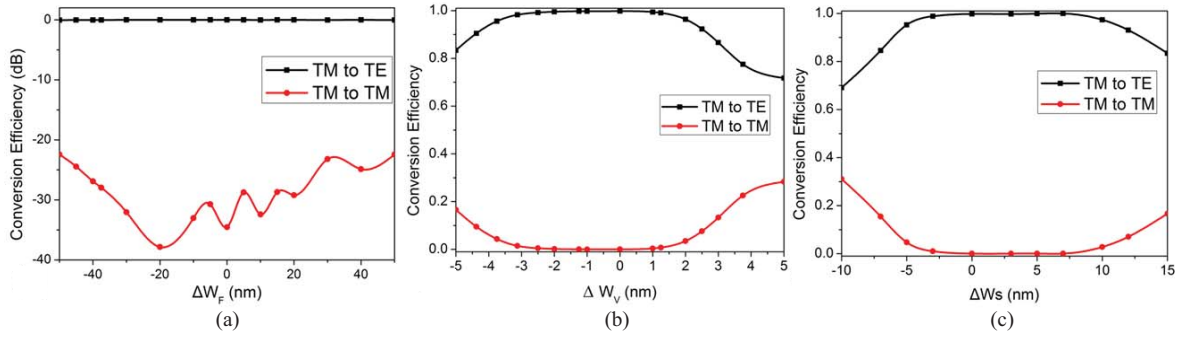


Fig. 6. Fabrication tolerances for the parameters of (a) W_F , (b) W_V , and (c) W_S .

from 1500 nm to 1600 nm and the conversion efficiency keeps more than 98% in this range.

III. FABRICATION TOLERANCE ANALYSIS

Considering the fabrication errors in the realistic fabrication process, we give an analysis on the tolerance of the key parameters, i.e. W_F , W_S and W_V (including W_0 , W_1 , W_2 , and W_3). The tolerance of W_F is not so critical, as is seen from Fig. 6a. When ΔW_F varies from -50 nm to 50 nm, the crosstalk is lower than -20 dB. Simulation results show that the conversion efficiency is more sensitive to both W_S and W_V . For the conversion efficiency of higher than 96%, ΔW_V must be controlled within the range from -3 to 2 nm, while ΔW_S from -5 to 10 nm. The physics underlying the fabrication tolerances difference is as follows. As W_F varies, the effective indices' convergence point is always around 270 nm, resulting in a relative large tolerance. On the other hand, in order to keep the convergence point around 270 nm, W_1 must be smaller than 270 nm and W_2 must be larger than 270 nm, which causes a tight tolerance of W_V . Meanwhile, the effective index of the slot waveguide is sensitive to the slot width W_S , thus resulting in a tight tolerance. Even though, the tolerance of W_V and W_S is still acceptable if electron beam lithography (EBL) technique [24] is used.

IV. CONCLUSION

We designed a novel polarization converter based on the slot tapered waveguide. The device can convert the first-order TM/TE mode to first-order TE/TM mode directly utilizing the principle of mode hybridization. After structural optimization, the efficiency of the converter can reach 99.78% at 1550 nm with a total length of 116 μm . Besides, the bandwidth for the conversion efficiency more than 98% is about 100 nm (1500 nm-1600 nm). The fabrication tolerance analysis demonstrates the feasibility of fabricating such a device.

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